LatticeMico32 On-Chip Memory Controller

The LatticeMico32 on-chip memory controller provides a slave interface to the WISHBONE bus master ports that allow them access to the Lattice Semiconductor FPGA embedded block RAMs (EBRs). The on-chip memory controller automatically instantiates the EBR using the parameterized module interface (PMI).

Features

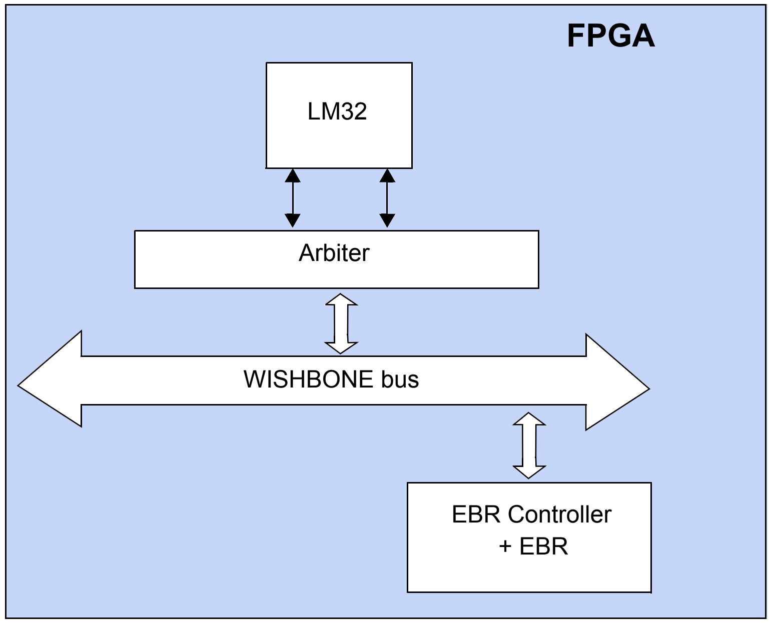
The LatticeMico32 on-chip memory controller includes the following features:

- WISHBONE B.3 interface
- Connector between WISHBONE master ports and Lattice Semiconductor EBR memory
- Classic WISHBONE write and read bus operations
- Support for 8-bit and 16-bit operations

As shown in Figure 1, the on-chip memory controller acts as a connector between the WISHBONE master ports and Lattice Semiconductor EBR memory. The master ports can write data to or read data from the EBR memory. When used, the memory controller instantiates the required EBRs.

For additional details about the WISHBONE bus, refer to the *LatticeMico32 Processor Reference Manual*.

Figure 1: LatticeMico32 On-Chip Memory Controller Slave Component



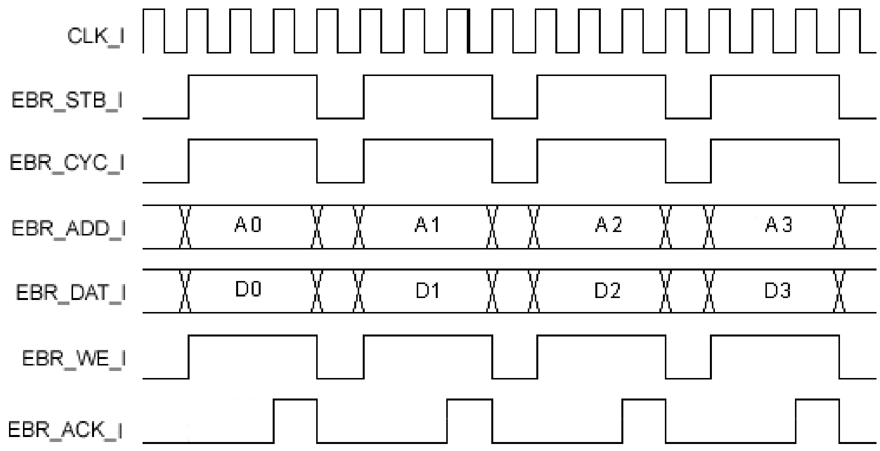
Functional Description

The LatticeMico32 on-chip memory controller provides classic WISHBONE write and read bus operations, each of which supports 8-bit, 16-bit, and 32-bit read and write operations. EBRs are included with the memory controller. The maximum memory size or address range is limited by the FPGA’s EBR block. The controller always completes the read and write in three cycles.

Figure 2 shows the port names and timing diagram of the on-chip memory controller in write mode.

The memory write occurs across three clock cycles. On the first clock cycle, the address is written from the controller to the memory. On the second clock cycle, the data is written from the controller to the memory. On the third clock cycle, the EBR_ACK_O signal is asserted.

Figure 2: On-Chip Memory Controller Timing Diagram for Write Mode



When the CTI signal value is 3'b000 and the WE signal is disabled, the master port reads one piece of data at a time, as shown in Figure 3. The width of the data transaction can be 8, 16, or 32 bits.

The memory read occurs across three clock cycles. On the first clock cycle, the read data address is written from the controller to the memory. On the second clock cycle, the data is returned to the controller from the memory. On the third clock cycle, the EBR_ACK_O signal is asserted, and the data is presented to the WISHBONE bus.

Figure 3: ON-Chip Memory Controller Timing Diagram for Read Mode

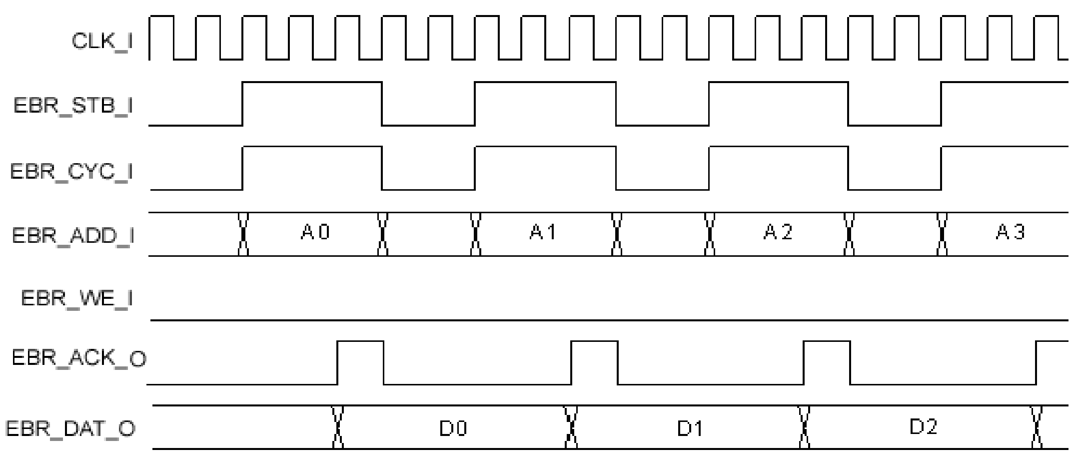


Figure 4 shows the port names and timing diagram of the on-chip memory controller in burst 4 write mode.

Figure 4: On-Chip Memory Controller Timing Diagram for Burst 4 Write Mode

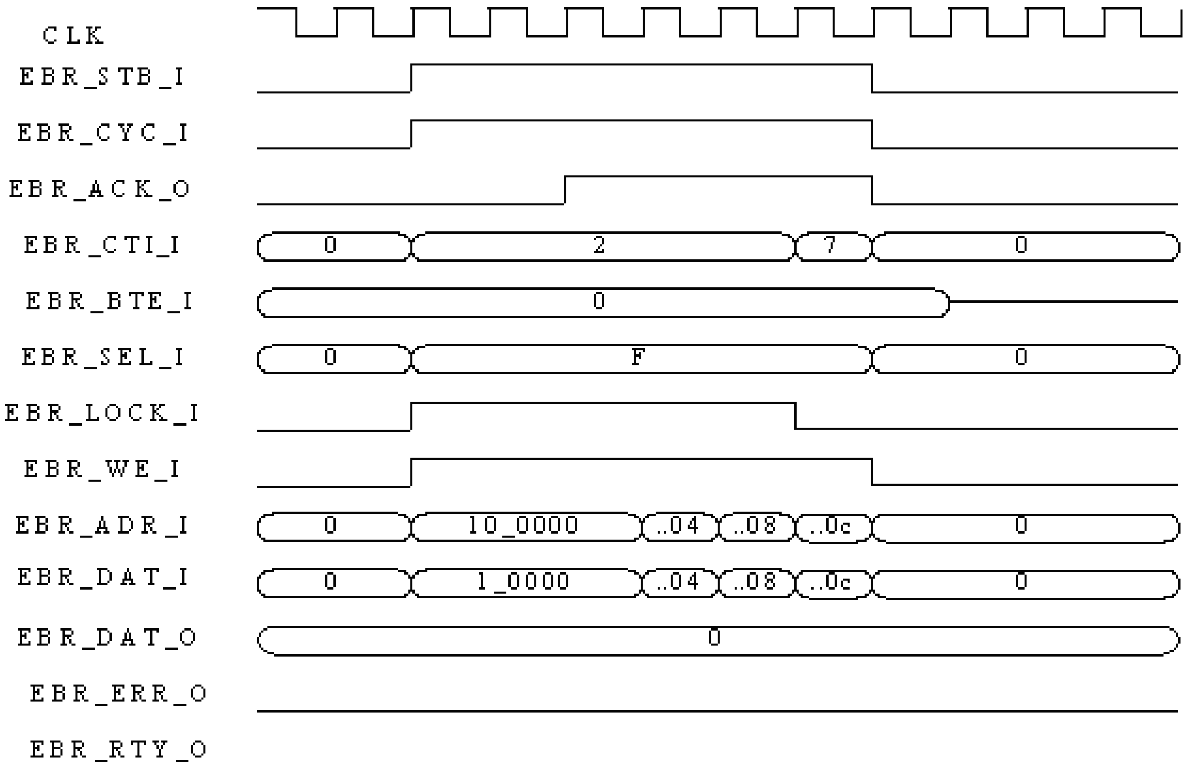
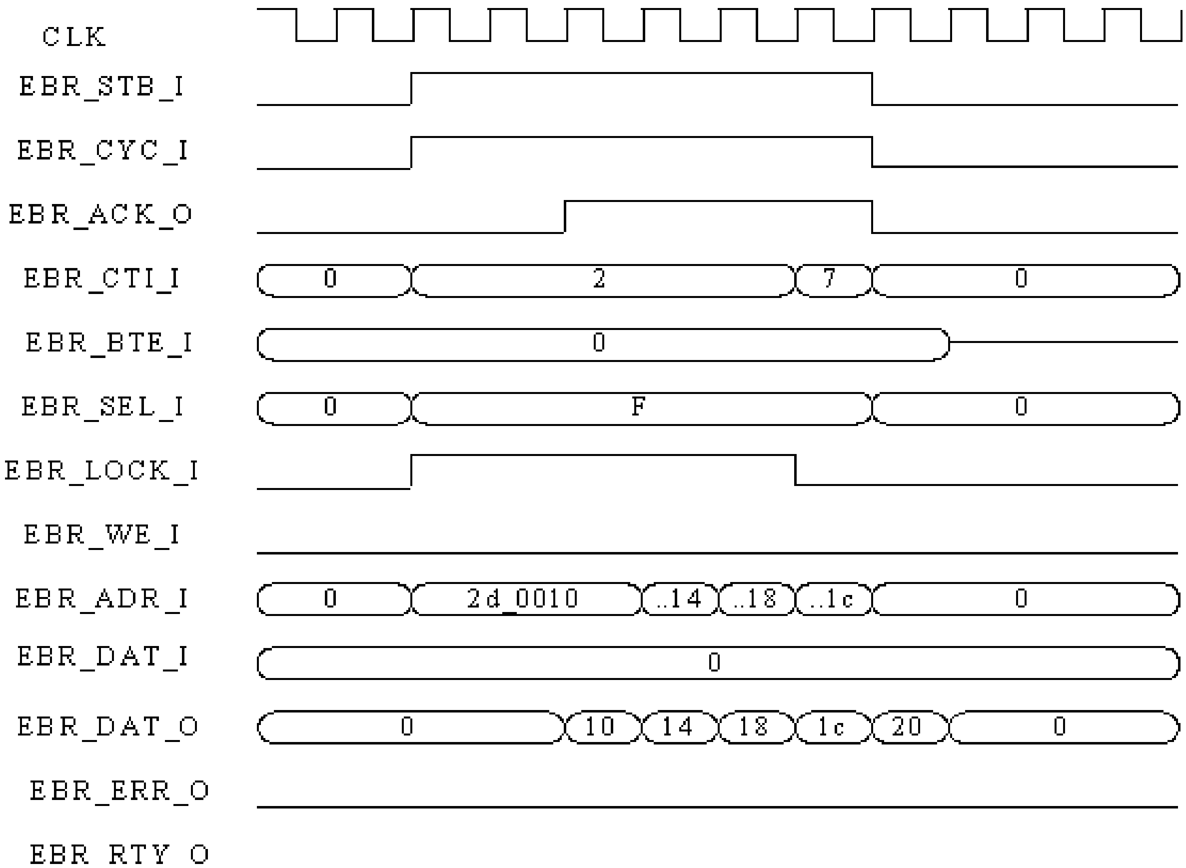


Figure 5 shows the port names and timing diagram of the on-chip memory controller in burst 4 read mode.

Figure 5: On-Chip Memory Controller Timing Diagram for Burst 4 Read Mode



Configuration

The following sections describe the graphical user interface (UI) parameters, the hardware description language (HDL) parameters, and the I/O ports that you can use to configure and operate the LatticeMico32 on-chip memory controller.

UI Parameters

Table 1 shows the UI parameters available for configuring the LatticeMico32 on-chip memory controller through the Mico System Builder (MSB) interface.

Table 1: On-Chip Memory Controller UI Parameters

Dialog Box Option	Description	Allowable Values	Default Value
Instance Name	Specifies the name of the on-chip memory controller instance.	alphanumeric and underscores	ebr
Base Address	Specifies the base address for the device. The minimum boundary alignment is 0X00. The base address can be configured automatically in the LatticeMico32 System builder.	0X00000000–0XFFFFFFFF If other components are included in the platform, the range of allowable values will vary.	0X00000000
Size of Memory (in bytes)	Specifies the total size (depth x width) of the memory implemented using embedded block RAMs (EBRs, in bits). The size is the limit of the EBR block in the selected FPGA.	0 – 72K	2048
Initialization File Name	Specifies the name of the memory initialization file. **Note**: The memory initialization file can be created from the software development environment. For a detailed description of the steps, refer to the “Deploying to On-Chip Memory” section of the *LatticeMico32 Software Developer User’s Guide*.	Valid file name	None
File Format	Specifies the format of the memory initialization file.	Hexadecimal, binary	Hexadecimal

Note: For LatticeEC/ECP devices, 1 EBR = 1 kb

Note

You can use the ispLEVER Memory Initialization Tool to change the content of initialized RAM (EBR) after synthesis, placement, and routing. For more information on the ispLEVER Memory Initialization Tool, choose **Help > Project Navigator > User Interface > Memory Initialization Dialog Box**.

HDL Parameters

Table 2 lists the parameters that appear in the HDL.

Table 2: On-Chip Memory Controller HDL Parameters

Parameter Name	Description	Allowable Values
BASE_ADDRESS	Specifies the base address for the device.	0X00000000–0XFFFFFFFF
SIZE	Specifies the total size of the memory, in bits (depth x width), implemented using embedded block RAMs (EBRs). The size is the limit of the EBR block in the selected FPGA.	0X00000000–0XFFFFFFFF
INIT_FILE_NAME	Specifies the name of the memory initialization file.	Valid file name
INIT_FILE_FORMAT	Specifies the format of the memory initialization file.	Hexadecimal, binary

I/O Ports

Table 3 describes the input and output ports of the LatticeMico32 on-chip memory controller.

Table 3: On-Chip Memory Controller I/O Ports

Signal Name	Active	Direction	Initial State	Description
CLK_I	HIGH	I	0	Input clock signal. The clock is rising edge active.
RST_I	HIGH	I	0	System reset signal
WISHBONE Slave Interface Signals				
EBR_ADDR_I [31:0]	—	I	0	Address input array. Address is generated by the master.
EBR_DAT_I[31:0]	—	I	0	Data input array, which is valid for a write request.
EBR_WE_I	—	I	0	Write enable signal. Value is 1 for a write and 0 for a read.
EBR_CYC_I	HIGH	I	0	Cycle input. When asserted, it indicates that a bus cycle is in progress.
EBR_STB_I	HIGH	I	0	Strobe input. When asserted, it indicates that the slave is selected.
EBR_SEL_I[3:0]	HIGH	I	0	Select input array, which indicates where the valid data is expected on the data bus. The read operation reads the bytes that are EBR_SEL high. The write operation writes only the selected bytes, leaving the others unchanged.
EBR_CTI_I[2:0]	—	I	0	Cycle-type indication signal
EBR_BTE_I[1:0]	—	I	0	Burst-type extension signal

Table 3: On-Chip Memory Controller I/O Ports (Continued)

Signal Name	Active	Direction	Initial State	Description
EBR_LOCK_I	HIGH	I	0	Slave bus locked
EBR_ACK_O	HIGH	O	0	Acknowledge output. When asserted, it indicates normal cycle termination.
EBR_DAT_O[31:0]	—	O	0	Data output array
EBR_ERR_O	HIGH	O	0	Slave error, which is always 0.
EBR_RTY_O	HIGH	O	0	Slave retry, which is always 0.

Software Support

The LatticeMico32 on-chip memory controller does not require associated software support. It can access a memory's location by treating it as a general-purpose read/write memory.